NALLABATI VENKATA DURGA SAI SARAT CHANDRA

saratchandranallabati@gmail.com · +91 6305761091 · linkedin.com/in/sarat-chandra-nallabati · sarat-2007.github.io · github.com/sarat-2007

EDUCATION

Amrita Vishwa Vidyapeetham, School of Engineering — Amritapuri, Kerala, India
B.Tech in Robotics and Artificial Intelligence Engineering — CGPA: 9.05 / 10.00 · Aug 2024 – May 2028 (3rd Year)
Relevant Coursework: Multivariable Calculus, Linear Algebra, PDEs, Robotics Kinematics & Dynamics, Microcontrollers & RTOS, Data Structures & Algorithms, Machine Learning, Computer Vision, Digital Signal Processing, Convex Optimization.

TECHNICAL SKILLS

Languages & Autonomy: Modern C++ (C++17/20), Python (Expert), Embedded C, SQL, ROS 2 (Jazzy/Humble), CycloneDDS, OpenCV, FreeRTOS
Embedded & Hardware: STM32 (ARM Cortex-M), Arduino, ESP32, Valetron A7670C 4G LTE, CAN 2.0B/CAN-FD, UART DMA, I2C, SPI, 240V Relays, KiCAD
Scientific ML & Edge AI: PINNs, Proper Orthogonal Decomposition (POD/SVD), SINDy, PyTorch, ANSYS Fluent CFD, AMD Ryzen AI NPU (XDNA)

PROFESSIONAL EXPERIENCE & SYSTEMS LEADERSHIP

CLUB MIRAI (Robotics & Embedded Systems Club) — Amrita Vishwa Vidyapeetham
Embedded Systems and Robotics Developer & Hardware Systems Lead — Aug 2023 – Present

- Engineered modular ROS 2 node architecture with CycloneDDS zero-copy middleware for real-time 50 Hz closed-loop control.
- Implemented bare-metal STM32 firmware utilizing DMA-driven circular UART buffers at 921,600 baud with zero packet drop.

KEY ENGINEERING & RESEARCH PROJECTS (4 FLAGSHIP SYSTEMS)

1. Evolution Edge: Self-Evolving Edge AI Routing Engine (AMD Developer Hackathon) — Team Lead · 2024
AMD Ryzen AI XDNA NPU, Vitis AI EP, ONNX Runtime, ROCm 6.x, PEFT/LoRA

- Engineered neuro-symbolic routing engine elevating local edge LLMs (Qwen/Llama) via real-time query complexity gating.
- Deployed quantized INT8 models to AMD Ryzen AI NPU via ONNX Runtime Vitis AI EP, achieving <28.4 ms latency within 8W TDP.

2. Physics-Informed Neural Network (PINN) Non-Newtonian Hemodynamics Pipeline — Lead Developer · 2024 – Present
PyTorch, SciPy, ANSYS Fluent CFD, Carreau-Yasuda Rheology, Adaptive NTK Balancing

- Formulated Navier-Stokes forward/inverse physics-loss PINN inverting blood rheology parameters from pulsatile CFD velocity fields.
- Implemented Neural Tangent Kernel (NTK) trace balancing, achieving <2.4% relative L_2 error against transient ANSYS Fluent CFD.

3. Reduced-Order Modeling (ROM) & Sparse Nonlinear Dynamics (SINDy) — Lead Developer · 2024
Proper Orthogonal Decomposition (POD/SVD), STLSQ Sparse Regression, Python, NumPy

- Constructed POD-Galerkin reduced basis capturing 99.2% turbulent kinetic energy across 12 modes with 1200× speedup.
- Extracted sparse nonlinear ODEs via L_1-regularized STLSQ across 15,000+ spatial mesh points with zero structural divergence.

4. Industrial GSM & 4G LTE Remote Automation System — Firmware Lead · 2024
Embedded C++, Arduino, A7670C 4G LTE/GSM Module, Non-Blocking AT Parser, 240V Relays

- Architected non-blocking AT command parser driving A7670C 4G LTE module with sender phone authentication and status telemetry.
- Hardened 4-channel opto-isolated 240V relay switching with RC snubber transient suppression under 90V–480V inductive surges.

PEER-REVIEWED PUBLICATIONS & PREPRINTS

1. Small Scale Raspberry Pi Based Autonomous Car: A Practical Guide to Understand Complexity — Accepted & Presented · Jan 2025
IEEE International Conference on Robotics and Mechatronics (ICRM 2025)

- Engineered real-time perception pipeline tracking lane boundaries at 35+ FPS on ARM Cortex-A SBC with closed-loop PID lateral steering.

2. 25-Day Coastal Field Performance of Solar Parabolic Dish Desalination System Under Tropical Climates — Co-Investigator · 2024
Elsevier Desalination Journal Submission · Empirical Thermodynamics

- Logged 116°C focal receiver temperature, achieved 99.5% TDS reduction (3,850 to 18 mg/L), and modeled correlation (Adjusted $R^2 = 0.83$).

HONORS & COMPETITIVE AWARDS

- Global Finalist (Top-5 Worldwide), IEEE IES GenAI Challenge (2024): Physics-guided diffusion models with thermodynamic Gibbs loss.
- ISRO YUVIKA Space Science Fellow (2020): Selected in Top 0.1% nationwide by Indian Space Research Organisation for space science training.
- National Anveshika Experimental Physics (NAEST 2021): Qualified 2 national screening tiers under Padma Shri Prof. H. C. Verma (IIT Kanpur).
- State Rank 497 / 150,000+, AP POLYCET (2022): Top 0.3% statewide engineering merit rank in competitive entrance exam.